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Research Trends in Agriculture Sciences

Volume - 16

Chief Editor

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Contents

Chapters		Page No.
1.	Necessary Nutrients of Plant: Nutrient Functions and Deficiency Symptoms *(Pradip Kumar Saini, Dr. R.K. Yadav, Anupam Singh, Dr. Alok Kumar Singh and Dr. R.N. Kewat)*	01-26
2.	Cropping Systems for Agro-Climatic Zones *(Krishna Murari Pandey and Madan Kumar Jha)*	27-40
3.	The Impact of Climate Change on Indian Agriculture *(Devegowda SR, Saket Kushwaha, Kalpana Kumari and Pavan MK)*	41-55
4.	Weather Forecasting in Agriculture *(Sankar T, Arul Prasad S, Sudarmanian N.S and Dheebakaran GA)*	57-81
5.	The Importance of Carbon and Water in Agriculture *(Sathish Mohan Botsa and Sunkara Jhansi Rani)*	83-97
6.	Organic Rice Production Technology *(Virendra Singh and Satybhan Singh)*	99-129
7.	Nutrient Management for Maize in Rice-Maize Relay Cropping System *(R Rex Immanuel)*	131-154

Chapter - 4
Weather Forecasting in Agriculture

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Chapter - 4

Weather Forecasting in Agriculture

Sankar T, Arul Prasad S, Sudarmanian N.S and Dheebakaran GA

Abstract

Agriculture is the most important sector of Indian Economy. Indian agriculture sector accounts for 18 per cent of India's gross domestic product (GDP) and provides employment to 50% of the countries workforce. Weather is an important natural basic for the life, which provides or destroys the opportunity for survival based on their magnitude. Weather forecasting is nothing but, the prediction of condition of atmosphere of a given specific area at a particular time scale by using the principles of physics, supplemented by a variety of statistical and empirical techniques and by technology. Short range weather forecast information's play a vital role in making short-term adjustments in daily agricultural operations thus helps to minimize losses from adverse weather conditions and increase the yield and quality of agricultural produce. Also, helps in long range or seasonal planning and the crops selection based on optimum climatic conditions. Seasonal forecast helps to improve the livelihoods of farmers in the region with high inter-annual rainfall variability. Successful weather prediction was performed since 1920's and practical use of Numerical Weather Forecast (NWF) model in use from the middle of 19th century. A number of forecast models, both global and regional are being used to create forecasts. The disaster caused by adverse weather condition brings a knowledge base among farmers in the form of Indigenous Technical Knowledge (ITK) that would help to protect them from vulnerability and disaster for possible events.

Keywords: weather forecast, short range forecast, seasonal forecast, numerical weather forecast, indigenous technical knowledge

Introduction

Agriculture is the most important sector of Indian Economy. Indian agriculture sector accounts for 18 per cent of India's gross domestic product (GDP) and provides employment to 50% of the countries workforce. India is

the world's largest producer of pulses, rice, wheat, spices and spice products. Weather is an important natural basic for the life, which provides or destroys the opportunity for survival based on their magnitude. Weather forecasting is nothing but, the prediction of condition of atmosphere of a given specific area at a particular time scale by using the principles of physics, supplemented by a variety of statistical and empirical techniques and by technology. Weather forecast is very much useful to save our human life and property, to protect the crops and it will tell us what to expect in our atmospheric environment.

Weather forecast information's play a vital role in making short-term adjustments in daily agricultural operations thus helps to minimize losses from adverse weather conditions and increase the yield and quality of agricultural produce. Also, helps in long range or seasonal planning and the crops selection based on optimum climatic conditions. Seasonal forecast helps to improve the livelihoods of farmers in the region with high inter-annual rainfall variability. Seasonal forecasts play an important role in deciding the economic policies of governments, during major drought was forecasted, monetary policy could be relaxed to maintain growth targets.

Weather forecast information's are used to know about the crop diversification strategies which help farmers to decide what crop to cultivate. High demand on weather forecasting is faced by agricultural community for their sustainable development. Accessing weather information is often boon to farmers and positively impacts the changing of farming practices. In older days, weather forecasting information were used for postharvest operations for example, light winds could help in the winnowing operations, wind energy conversion, scheduling of a power system and the dynamic control of the wind turbine to avoid unnecessary cost involved in harvesting and using of pesticide and fertilizers to increase the yield.

Successful weather prediction was performed since 1920's and practical use of Numerical Weather Forecast (NWF) model in use from the middle of 19th century. A number of forecast models, both global and regional are being used to create forecasts. Climate forecast using empirical studies helps the farmers to reduce the weather risk and vulnerability during extreme climates and drought and helps to maximize the opportunities when favourable climatic conditions were predicted.

Among the required forecast, rainfall is the most important one, which controls the crop production over a particular region and finally the country's economy. Moisture conservation planning has to be taken under

weak monsoon period and flood relief under strong monsoon periods has to be taken mostly based on the prevailing weather condition on a particular area.

Special weather forecast is issued during the abnormal weather events such as cyclone, drought, flood, cold or heat wave and forest fires. They do not have any specific code and duration. Advanced tools, techniques and models viz., Radars and satellite are being used for the prediction and continuous tracking of the events when the system comes within radar range. It provides the additional information to find out the wind distribution, convective cloud cluster, rainfall rate etc. with help of Doppler Weather Radar (DWRs) and also useful for cyclone monitoring and prediction.

Indigenous Technical Knowledge (ITK) otherwise called as traditional knowledge (TK), indigenous knowledge (IK), and local knowledge and usually refers to "knowledge system embedded in the cultural tradition of regional, indigenous, or local community". The disaster caused by adverse weather condition brings a knowledge base among farmers in the form of Indigenous Technical Knowledge (ITK) that would help to protect them from vulnerability and disaster for possible events.

Astrometeorology is nothing but the science dealing with the interaction between planetary movement and weather and even today it is still practiced by few. Surprisingly, Village astrologers (*pandits*) are right in their weather predictions with high percentage of accuracy.

Objective

Objective of this chapter is to provide information about forecast techniques used in weather prediction and preparation of agro-advisories that boost agricultural.

4.1 Weather forecast

Weather forecasting is nothing but, the prediction of condition of atmosphere of a given specific area at a particular time scale by using the principles of physics, supplemented by a variety of statistical and empirical techniques and by technology. Weather forecast is very much useful to save our human life and property, to protect the crops and it will tell us what to expect in our atmospheric environment.

Aristotle, the first ancient Greek scholar who used the science for preparing weather forecast during (384-322BC). The interaction of four elements i.e. earth, fire, air and water causes weather and it was mentioned in a book called 'Meteorological' written by Aristotle. From his work the term

‘Meteorology’ was derived. Weather observations have been observed since the start of the civilization. Visual as well as sensual weather phenomena had been recorded till 18th century. Until the invention of various meteorological instruments such type of observations were continued for a longer period.

4.1.1 Procedures of weather forecasting

Now a day, the weather forecasting has become a science and it is performed by adopting the following procedures (steps):

Weather data recordings (temperature, pressure, wind speed and direction, cloud formation, humidity level and precipitation, visibility, storms etc.)

↓

Weather data collection from weather recording (observations centers) stations scattered world over including both land and ocean surfaces

↓

Transmission of weather data collected from major weather stations to sub centers.

↓

Compilation of weather data

↓

Weather data plotting on maps and daily weather records, synoptic charts etc

↓

Analyzing of weather charts and maps with the help of electronic computers etc

↓

Final forecasting of weather and numerical modeling

4.1.2 Different weather forecasting services

Ancient people have made attempt to forecast the weather informally for millennia and formally since the 19th century. Indian farming community is in great need of accessing weather information’s to manage and make a plan of cropping operations and their livelihood. Weather forecasting is necessary in forest management to prevent and control wildfires and for predicting conditions for the development of harmful insects (Figure 1).

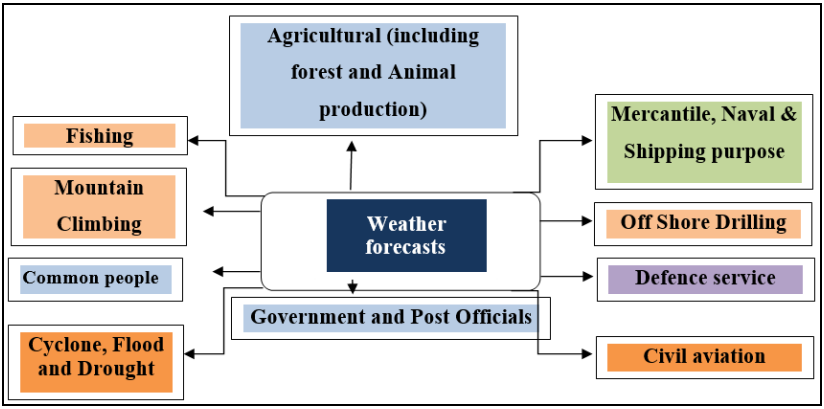


Fig 1

4.2 Need or forecast importance

Weather is having many cultural, economic & social impacts in the society. Weather alone influences the crop production upto 50%. Among the required forecast, rainfall is the most important one, which controls the crop production over a particular region and finally the country's economy. Moisture conservation planning has to be taken under weak monsoon period and flood relief under strong monsoon periods has to be taken mostly based on the prevailing weather condition on a particular area. We can reduce the damages caused either directly or indirectly by unfavorable weather condition by utilizing proper weather forecasts. The frequent crop damages can be reduced if appropriate timely forecast information on incidence of pest and diseases is given based on weather parameters. It helps to hold fluctuations in the prices of food grains through checking of buffer stock operations. That means, prices fall in good quality monsoon years, the government has taking steps to purchase agricultural commodities and in bad years when price tend to increase, it may discharge a part of what it had purchased. Usage of water can be planned over a region depending on the forecast level. The forecast of the weather events help the farmers for planning their farm operations thus, it helps to increase their farm income.

Weather forecasting helps in following farm operation

- It helps to undertake or withheld the sowing operation
- To scheduling of irrigation to crops or not
- To timely application of fertilizers
- To plan complete harvesting or to withhold it

- It also helps to take measures against frost damages
- It helps to store and transport of food grains
- It helps to manage the inter-cultural operations like plugging harrowing hoeing etc
- It helps to take necessary actions to protect livestock against adverse weather conditions

4.3 Agencies

ICAR made an attempt to provide valuable agro-met information by considering the growing acceptance of farmers for the internet and mobile based services, to the users through a common platform 'Crop Weather Outlook'. Under the All India Coordinated Research Project on Agrometeorology (AICRPAM) highlighted Agrometeorological information is generated by using this platform and its Collaborating and Cooperating Centres along with "Value Added Agro-advisory Reports". This website provide information's on the research activities conducted by the Agro-met Centres located at various State Agricultural Universities as well as, also linking the other associated web sites so that the observer can access all the relevant information based on his timely needs and efficiently. This website is operated by Central Research Institute for Dry land Agriculture (CRIDA), Hyderabad. Key information's available from this site are daily weather report, monsoon status, districts of deficit rainfall.

India meteorological department (IMD) is started in 1932 at Pune; this is the agency in India responsible for issuing weather forecast in the country. It operates to serve the agricultural community through different types of forecasting. India has been divided into 36 meteorological subdivisions based on homogeneity in rainfall distribution. Forecast is issued meteorological subdivision wise and the same is communicated to the users. IMD has 6 Regional Meteorological Centers (RMC) to give data for preparing forecasts. These centers are Mumbai, Nagpur, Chennai, New Delhi, Guwahati and Kolkata.

National Centre for Medium Range Weather Forecast (NCMRWF) was established by the Department of Science and Technology in 1988. Development of operational medium range forecast and issue of Agrometeorology Advisory service for 127 Agro Climatic Zones are in collaboration with State Agricultural Universities. Medium Range Forecast is issued by NCMRWF once or twice a week, covering a period of 3 to 5 days, are utilized by AMFU located in SAUs for preparing weather-based agro advisories. From June 2008 the India Meteorological Department has

reorganized and initiated to provide district level forecast now generated by respective Regional Meteorological Centre.

4.4 Types of weather forecasting

On the basis of duration of the validity, the forecasts are classified into 6 categories,

1. Now-casting
2. Very short ranges forecast
3. Short ranges forecast
4. Medium ranges forecast
5. Extended ranges forecast
6. Long ranges forecast

4.4.1 Now-casting

Now-casting comprises the complete report of instant weather condition with its specific forecast obtained by extrapolation for a period of 0 to 6 hours ahead and with this time limit, maximum possibility for forecasting smaller features such as individual storms with realistic accuracies. Auto Now-casting System was developed by National Centre for Atmospheric Research (NCAR) that combines radar, satellite and surface data to forecast for next few hours. It is the most helpful tools to provide warnings the public from adverse effect, high weather impact including tropical cyclone, thunderstorm and tornado which causes flash flood, lightning strike and destructive wind.

Now-casting contributes to the

- To reduce the losses and injuries due to adverse weather hazard
- To reduce the damage of private, public, and industrial, property
- To improve effectiveness and savings for industry, transportation and agriculture

4.4.2 Very short range forecast

Very short range forecast provides information for the period upto twelve hours. Accuracy of these forecast is very high and the potential focus is to moderate the weather conditions. It is closely related to local weather phenomenon like thunder storms, cyclone, cold waves and heat waves etc.

4.4.3 Short-range weather forecasts

Weather information is given for a period from 12 to 72 hours is known as short range forecasts. Short range forecast focus is to the farmers for

adjusting the schedule of irrigation, fertilizer application, foliar spray and harvest. These forecast is comprises forecast and extreme weather events warning to agriculture with valid for 36 hours thus, it includes an outlook for next 2 days on basis of synoptic conditions. The forecast contains information on cloud, rainfall, maximum and minimum temperatures, abnormal heat and cold waves, low pressure, cyclone warning, frost and wind speed of particular time. The forecasts are issued by the analyzing the maps, weather charts, synoptic charts, satellite images. Short-range weather forecast determines the change of atmospheric weather of a particular location. It is commonly used in transportation and by fishermen. SRWF is for taking preventive measures against adverse weather, the reaction time to the farmers is too short and accuracy of weather forecast is around 70-80 percent. Also, it is useful in weather based agricultural operations thus, helps to reduce the production costs, reduce the wastage of planting, pest control, avoid more irrigation, harvesting, storage loss and marketing.

4.4.4 Medium range of weather forecasts

Medium ranges of weather forecast are given for 3 to 10 days. During 1988, Government of India established the centre to develop a medium range of weather forecast (The National centre for Medium Range Weather Forecasting, NCMRWF) in New Delhi and this is the hub which disseminates information on appropriate time. MRWF is also given by India Meteorological Department (IMD) on various parameters viz rainfall amount, minimum and maximum temperature, windspeed and cloud amount for all over india. Enormous numerical computations are requiring forecasting in weather science. Since 2011, Tamil Nadu Agricultural University is also proving medium range weather forecast at block level with a lead time of six days. MRF is used to manage the labour and farm equipment, to decide sowing and harvest time, to fix the sowing depth of seeds and to protect the livestock against heat and cold wave conditions. Presently, TNAU forecast accuracy of medium range forecast is 60 to 70 per cent at district level and 70 to 80 per cent at block level. Forecast prediction is done by calculating the average of past and present weather condition.

4.4.5 Extended-range weather forecasting

This type of weather forecasting is given more than 10 day and upto 30 days. Here, various weather parameter, generally taken average and express as a departure from climate values for that specific time period. This is involves in farm planing, input and out put management at farm and policy level.

4.4.6 Long range of weather forecasting

Long range of weather forecasts are covers a time span from a fortnight to a season of the year. These are basically statements and the exact accuracy is low. There are three types in long range weather forecasting.

i) **Monthly forecast:** Averaged weather parameters of a month and express as a departure (deviation, variation, and anomaly) from climate values (not necessarily the coming month)

ii) **Three months or 90 days forecast:** Averaged weather parameters of 90 days or three months and express as a departure from climate values for that 90 days period (not necessarily the coming 90 days time period)

iii) **Seasonal forecast:** Average weather parameters of season and express as a departure from climate values for that specific season

4.5 Methods of weather forecasting

On the basis of analyzing, there are 6 weather forecasting methods such as

1. Synoptic method
2. Numerical prediction method
3. Statistical method
4. Analogue method
5. Persistence Forecasting
6. Climatological approach method

4.5.1 Synoptic Weather Forecast

Synoptic forecasting means the observations made on a set of meteorological variables viz. rainfall, wind and pressure systems, maximum and minimum temperatures are measured simultaneously at various locations at specific times. Using these observations, surface and upper air charts are prepared which give the present state of atmosphere. The inference on expected movement of weather systems are drawn using the previous and present charts. This is mainly used for preparing short range weather forecasts.

4.5.1.1 Synoptic chart

A synoptic chart is scientifically known as weather map of a particular area. Synoptic charts are providing information on distributions, movements and pattern of air pressure, rainfall amount, wind speed and temperature (different synoptic hours are plotted i.e., 00, 03, 06, 09, 12, 15, 18, 21 UTC). Those informations are conveyed through usage of symbol, which could be

explained in a legend. This synoptic chart is used to disseminate information of the instant weather and to forecast future weather patterns.

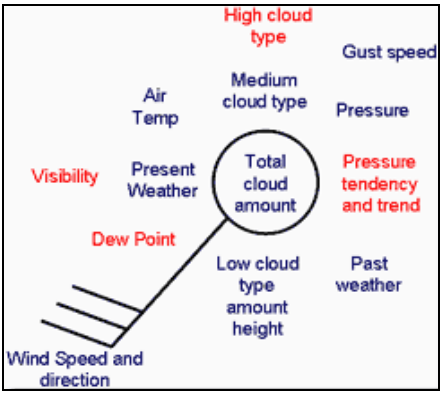


Fig 2

4.5.1.1.1 Synoptic chart-Pressure systems

The most significant characteristic in a synoptic chart is the fine black lines called isobars (Figure 3 & 4). Isobars are the line which connects the equal points of same atmospheric (air) pressure (Figure 3). The instrument used to measure air pressure in hecto Pascals (hPa) is Barometer. The average air atmospheric pressure value at sea level is 1013 hPa. The measurements above mean global atmospheric pressure are called as high pressure system and are taken as an area of sinking air. In high pressure system areas generally the weather is calm and fine. In maps, these higher pressures are often marked by an 'H' symbol. These systems are otherwise known as anticyclones. Value below this atmospheric pressure is called a low pressure system and is known as an area of rising air. In low pressure system areas are mostly in unsettled weather conditions, which can cause rains. In maps, it is depicted by using 'L' symbol. These systems are otherwise called as cyclones.

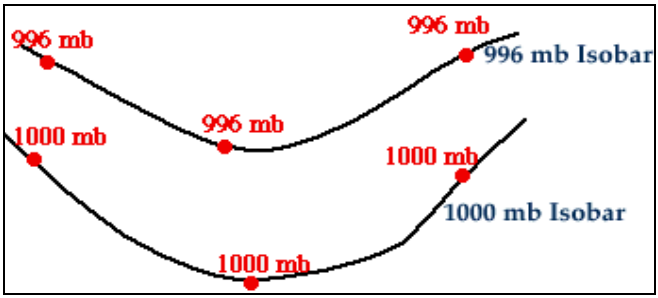


Fig 3

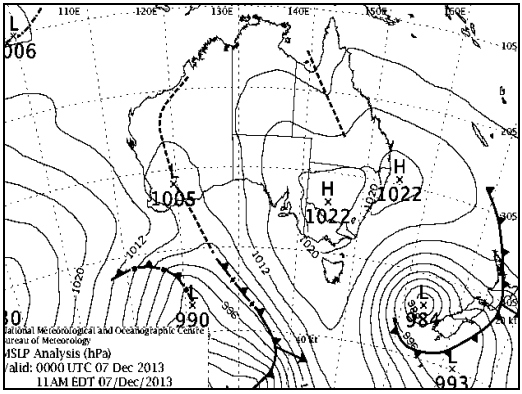


Fig 4

4.5.1.1.2 Synoptic charts-precipitation

Precipitation means including a forms of hail, snow and dew. However, rainfall is the most general form of precipitation received and it is a most common feature used on synoptic charts, because it has a significant role on the daily human activities. Using Shading rainfall is usually showing in synoptic charts (Figure 5). By using legends these shadings are explained in synoptic chart as representing the areas which receives rainfall amount in last 24 hours. Good indicators of rains are low pressure systems, cold and warm fronts and total cloud covers over the sky are denoted as Octa (from 0 to 9 Octa) (Figure 6).

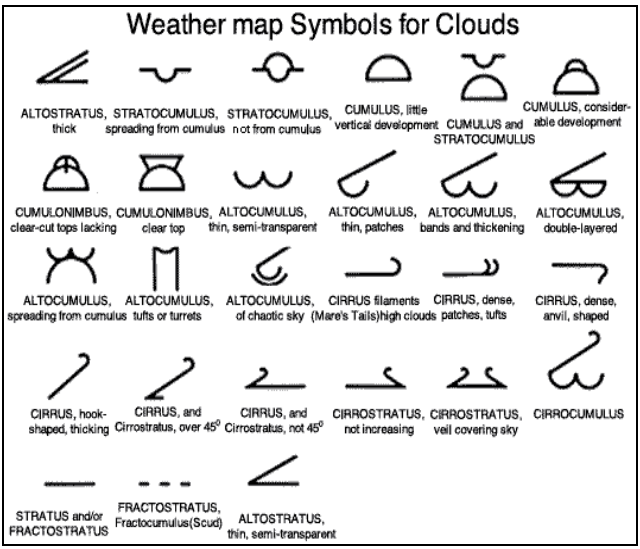


Fig 5

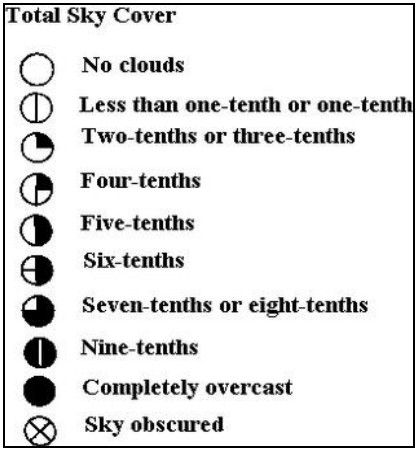


Fig 6

4.5.1.1.3 Synoptic chart-wind

Horizontal movement of air is generally called as wind. Using a Wind Vane one can measure wind direction, the device use to turn its axis towards pointing the direction of the wind. Wind speed can be measured by using an anemometer, the device use to rotate its cups based on pressure differences to determine wind speed. By using wind barb wind speed and direction are indicated on synoptic charts (a dot with a straight line attached) (Figure 7). The direction of barb stem points is indicates the direction of the wind is coming from (Figure 8). The lines are added to the barb if, an increase in wind speed and it makes the appearances like an arrow. The barb contains huge vanes, it indicates the speed of wind is high.

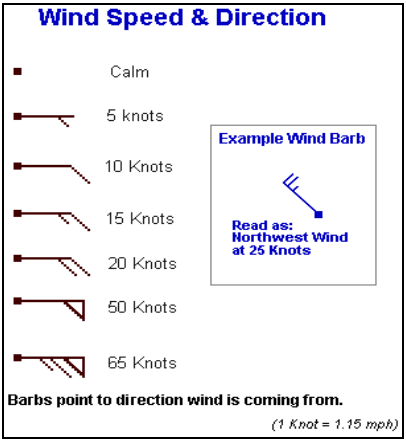


Fig 7

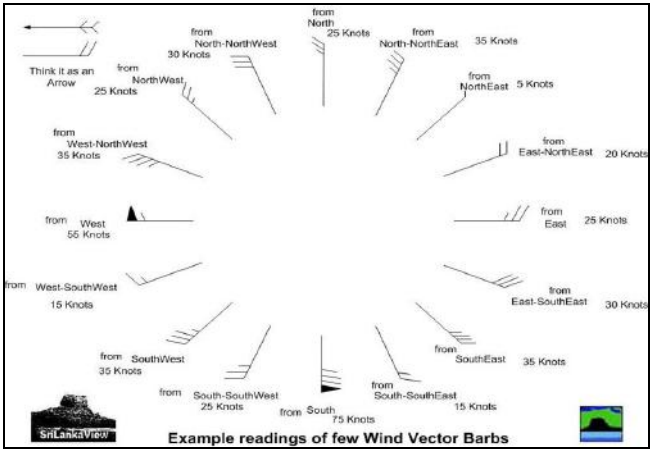


Fig 8

4.5.1.1.4 Synoptic chart-temperature

A synoptic chart doesn't have any features, any numerical or symbolic measurements of temperature (degrees of hotness or coldness). Temperature readings and predicted forecasts are to be displayed in an accompanying table on synoptic charts where they can be easily recognized and interpreted by the general public. Other symbols in synoptic charts are depicted in Figure 9.

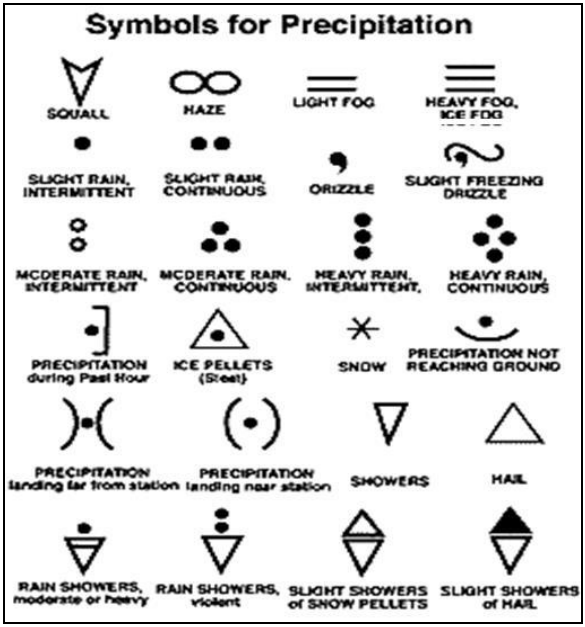


Fig 9

4.8.2 Numerical Weather Prediction method

Numerical Weather Prediction (NWP) can use computer powers to develop a forecast. It was first attempted in the 1920. Forecast models, with complex computer programs used, and it is run with help of super-computer and provides forecast on many atmospheric variables i.e., temperature, pressure, wind, and rainfall. A predictor examines how the future is predicted by the computer that will interact to produce the day's weather.

Effect of all forces viz. coriolis force, pressure gradient force, gravitational force, centrifugal and frictional forces etc that influence the motion of atmosphere is expressed in seven equations based on certain laws of physics. The integrated form of these equations is known as atmospheric model. Atmospheric models can be used for predicting the final state of weather through purely mathematical process taking into account the current state of weather in a number of synoptic stations of a given region. The NWP technique is used in medium range weather forecasting.

4.5.3 Statistical weather forecast

Statistical forecasts are building up of weather forecasts through purely statistical means. Mostly based on multiple regression equation and statistical methods are necessary because the atmosphere is a non-linear dynamical in nature. Statistical methods are used in long range and seasonal forecasts. IMD was using Auto Regressive Integrated Moving Average (ARIMA) models for predicting Indian monsoon rainfall based on 16 global-land-ocean atmospheric variables. Using the technique total rainfall quantity during the southwest monsoon (June to September) was predicted well in advance on 25th May every year by IMD. A new 8-10 parameter Power Regression model is devised in place of 16 parameter model since 2003 onwards for estimating the monsoon rainfall at national level. It helps the IMD to advance the Long Range Forecast by 40 days and generate a quantitative forecast update by mid July, which will be useful for contingency planning.

The procedure is to compile the statistical data related to temperature, wind velocity and direction, amount of cloudiness, humidity, and to the specific season of the year. This chart provides information of the maximum temperature for the day from the data of the current conditions. They are the projections of the rainfall and temperatures that may be expected during these periods. This will provide an idea of whether the temperature and rainfall in the region will be above or below the normal.

4.5.4 Analogue forecast

Analogue forecast are investigative today's forecast scenario and remembering a day in the past when the weather scenario looks very similar (an analog) and the interpreter will predict whether, the forecast would perform the same as it did in the past. This is most tedious technique to apply as there is rarely a perfect analog for an event in the future. This method of forecasting is also known as pattern recognition.

4.5.5 Persistence forecasting

Persistence forecasting is assumed that the current and future weather will be the same and there would be no changes in the weather conditions. Persistence forecasts are generally outstanding only for short periods (few hours) and the accuracy would reduce as the time period lengthens. For an example, suppose it is raining now, a forecast of rain for another three hours will be a persistence forecast.

4.5.6 Climatological approach

Climatological method of approach is a simple way to produce valid forecast. This method comprises computation and averaging of past weather statistic data over many years to formulate valid forecasts. An example, if we are used the climatological method to forecast the weather for Coimbatore on July 4th, we have to go-through all the weather data that has been recorded from July 4th of previous year and take an average of it. If we are going to make a forecast for temperature and precipitation, the recorded weather data for temperature and precipitation over past years at Coimbatore is used to compute the averages.

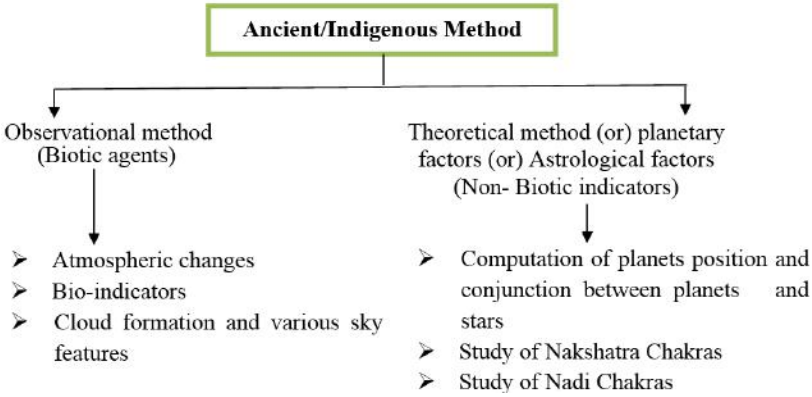
4.6 Methods of weather forecasts-indigenous technical knowledge (ITK)

Indigenous Technical Knowledge (ITK) otherwise called as traditional knowledge (TK), indigenous knowledge (IK), and local knowledge and usually refers to “knowledge system embedded in the cultural tradition of regional, indigenous, or local community”. The disaster caused by adverse weather condition brings a knowledge base among farmers in the form of Indigenous Technical Knowledge (ITK) that would help to protect them from vulnerability and disaster for possible events. We cannot replace the local knowledge system with scientific knowledge, because they are holistic and farmers to decide with the ability of their own and prepare psychologically for the coming agricultural year.

4.6.1 Importance of ITK

Various types of indicator and predictor (environmental, biological, and traditional believes) are used through ITK by the farmer to make farming

decision and protective measure. It is always based on locally condition and needs and is always dynamic in nurture by means of observation. It should be significant to record this valuable knowledge before it become “extinct”. Rural population successfully relied on indigenous forecasting method to plan their agricultural activities efficiently and effectively.



4.6.2 Rankings occupied by the indicators in rain forecasting-order of magnitude

1) Clouds (16.7%)
2) Wind (10 %)
3) Bird and Animals (8.80%)
4) Lightening (7.80 %)
5) Livestock (7.30 %)
6) Moon (6.70 %)
7) Trees (5.60 %)
8) Sky (5.6%)
9) Rainbows (5.0 %)
10) Termites and Ants (5.0 %)
11) Dragon fly, mosquitoes and houseflies (4.40%)
12) Food item (4.40%)
13) Stars (3.90 %)

4.6.3 ITK Vs cloud, sky, rainbow

If the Cloud appearances are in black and thick and it is perpendicular to the orbit of the sun in the morning, rain is approaching. Suppose, sky is fully covered with reddish coloured clouds after a long rainfall occurrence thus, results in failure of rain again in that particular season. In cyclone period cloud tends to move in a group from east to west, we can be expecting rain in next two days. Appearance of red sky at South west direction results in rain within 18 days. As well dark black color clouds are in night without stars, the possibility of rain will occur on that night. Rainbow on the western side is the indication of occurrence of good rainfall. Whereas, in eastern side is the indication of drought occurrence.

4.6.4 ITK Vs wind, roaring sea, halo around moon

In Dry season, winds blow towards the ocean to pick up water and returns onset of the rainy season. Westerly winds bring heat and south westerly bring coolness and clouds and occurrence of rains in monsoon season whereas northeast wind brings rain. Northwest (sanimuzhai) winds are wiped out the rain. South or north east winds (female wind) are mobilizing the cloud and South west winds (male wind) condense it and produce rainfall. Salts are released from sea due to interaction effect of sunlight and air that salts are acting as condensation nuclei and produce rains. The roaring of sea and the possibility of rain occurrence are positively correlated. If presence of the circles around the moon as bigger size, chance of getting less rainfall. Whereas, it is vice-versa in appearance of smaller circles and expect more rains for next 2 days (Figure 10 & 11).

Fig 10

Fig 11

4.6.5 ITK Vs birds and animals

Tiny insects swept up high on warm thermal region suppose air pressure is predicted high and the swallows to head upwards. Whereas, Air pressure is very low, occurrence of rainfall is more and swallows will find tasty feed around closer to the ground. Termites are appeared before the onset of rainfall and it will disappear after rainfall stop. Increasing moisture content can cause restlessness, sweating and flapping of ears by goats (Figure 12). Cattle can predict rain for next few hours earlier through sense cool breeze which is developed before rain for a short period. When humidity reaches goats jump joyfully (Figure 13). Male bamboo partridges are roaring frequently during spring and summer seasons in the morning after sunrise, rain is expected in the immediate future. Roaring of the bamboo partridge at the time of raining period in the morning is the indication of stopping rain (Figure 14).

Fig 12 **Fig 13**

Fig 14

4.6.6 ITK Vs flowering pattern

During the summer season full blooms are observed in peach, plum, and neem trees (Figure 15). Cassia fistula gives full blooms 45 days before onset of monsoon. In Ficus and Mango trees bears flowers and production of new leaves are noticed before onset of rainfall (Figure 16). Flowering of bamboo trees is an indication of upcoming drought. If, Bamboo bears flowers we can expect famine in the coming years (Figure 17).

Fig 15 **Fig 16** **Fig 17**

4.6.7 Prediction from panchangs

The nakshatra period is mostly based on the solar calendar. The basis for rain prediction in Panchangs are sun’s entrance time along with the combinations of other nakshatras (27) with specific zodiac signs (12 in number) to predict the quantity and rainy periods in particular year and the

changing in weather are related with movement of the sun, the moon and seven other planets in certain positions and act alone or in combination.

4.7 Forecast verification

The forecasts quality assessing process is called as forecast verification. Otherwise, Verification is the method of comparing forecast to appropriate observation. Verification is one feature of measuring forecast goodness. Forecasters are used a more appropriate term “evaluation” for many other purposes. The verification data set contains a collection of forecast or observation pairs whose joint behavior can be characterized in terms of the relative frequencies of the possible combinations of forecast or observation outcomes.

Forecast $(Y_i) = Y_1, Y_2 \ldots Y_i$

Observation $(O_j) = O_1, O_2 \ldots O_j$

The combine distribution of the forecasts and observations are given as,

$P(Y_i, O_j) = Pr\{Y_i, O_j\} = Pr\{Y_i \cap O_j\}$; $i = 1, 2 \ldots I$; $J = 1, 2 \ldots J$

4.7.1 Attributes of forecast performance

In forecast verification, reliability and resolution are the two main attributes of forecast systems. These attributes statistically related the performance of a forecast system to verifying data in an abstract sense. Forecast attributes are separately defined in the literature for systems that generate forecasts of particular formats or types.

Accuracy	Average correspondence between individual forecasts and the events predicted by observer.
Bias	The correspondence between the average forecast and the average observed value of the predicted.
Reliability	Pertains to the reliability of the forecast to the average observation, for specific values of the forecast.
Resolution	The degree to which the forecasts sort the observed events into groups that is different from each other.
Discrimination	Converse of resolution, pertains to differences between the conditional averages of the forecast for different values of the observation.
Sharpness	Characterize the unconditional distribution (relative frequencies of use) of the forecast.
Calibration	The aim of calibration is to make the form for each class of forecasts statistically more consistent with the distribution of the corresponding verifying observations.
Value	The value of forecast systems is assessed for the users (both resolution and reliability are considered)

4.7.2 Purpose of forecast verification

i. Administrative purposes are monitoring performance and choice of model or model configuration

ii. Scientific purposes are identifying and correcting model flaws; forecast improvement

iii. Economic purposes are to improve decision makings and "Feeding" decision models or decision support systems

4.7.3 Verification steps

Identifying multiple *verification attributes* that can answer to the questions of interest

⇩

Selection of *measures and graphics* that appropriately helps to measure and represent the attributes of interest

⇩

Identifying a *standard of comparison* that provides a reference level of skill (e.g., persistence, climatology, old model)

4.8 Special forecast for natural calamities

4.8.1 Drought forecast

Drought is "nothing but a prolonged situation of abnormally low rainfall, leading to water shortage over a period". The long range forecast issued by IMD is found reasonably well except in few years. Also, information given on region wise rainfall is undependable. The short and medium range forecast of rainfall may be useful for assessing droughts. But the accuracy of location specific weather forecasts is in question and response time of farmers is very limited to combat drought. Statistically, based on observed historical data, we can correlate between global variables and regional weather patterns. This is easy to calculate as this does not require "complicated" physical models.

4.8.2 Flood forecast

Flood is "a condition of high water level along a river channel or on the coast that leads to inundation of land, which is not usually submerged". Flood Forecast & Warning Organization was set up in CWC in 1969 to establish forecasting sites on inter-state rivers at various flood prone places in the country. Later, 41 forecasting sites were added in 1969. Extension of the service followed from time to time and now the river forecasting has been expanded over the years to cover nine major inter-state flood prone

river basins, which includes of 71 sub-river basins traversing the country. As on today the number of flood forecast sites has been increased to 175 (Figure 18).

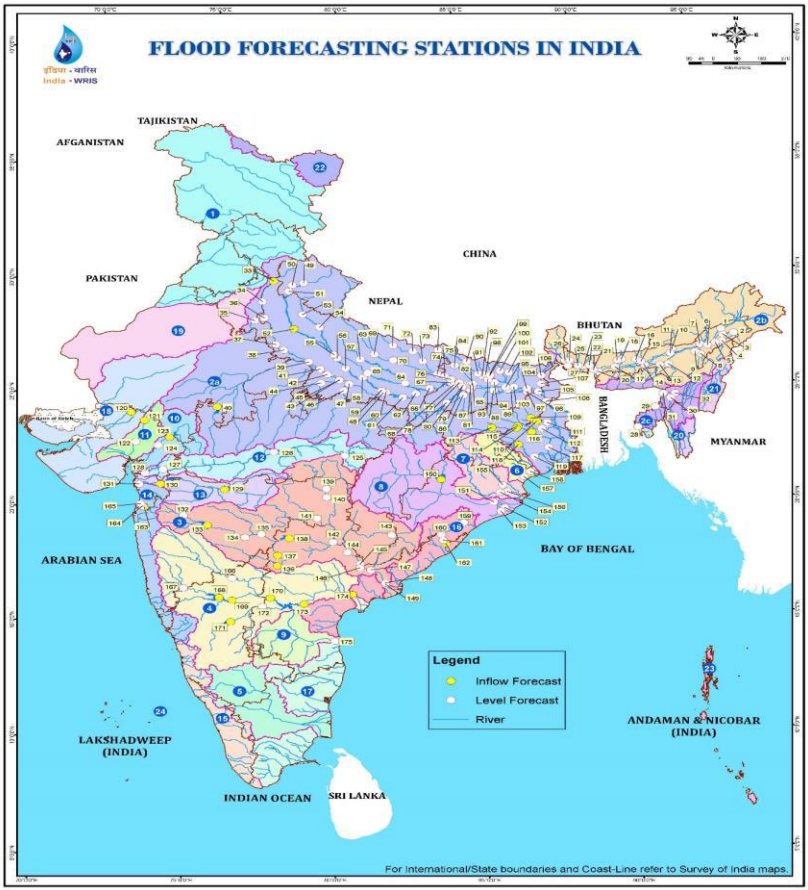


Fig 18

4.8.3 Heat waves forecast

A heat wave is nothing but "a prolonged period of excessively hot weather, which may be accompanied by high humidity, especially in oceanic climate countries". A heat wave is measured relative to the usual weather in the area and normal temperatures for the season.

4.8.3.1 Heat wave criteria by IMD

Heat wave is ignored where normally maximum temperature of a station reaches at least 40 ºC for Plains and at least 30 ºC for Hilly regions. When

normal maximum temperature of a station is less than or equal to 40 ºC, increase from normal is 5 ºC to 6 ºC is called as heat wave and the increase from normal is 7 ºC or more is called as severe heat wave. When normal maximum temperature of a station is more than 40 ºC, increase from normal is 4 ºC to 5 ºC is called as heat wave and the increasing from normal is 6 ºC or more is called as severe heat wave. Heat wave could be declared, when actual maximum temperature still 45 ºC or more, irrespective of normal maximum temperature.

4.8.4 Cyclone forecast

A tropical cyclone is a intense low pressure system along with very strong winds circulating around in anticlockwise in the northern hemisphere and clockwise in the southern hemisphere with a diameter varying between 150 to 1000 km approximately.

4.8.4.1 IMD four stage warning system

1. **Pre cyclone watch:** Issued by IMD HQ. This bulletin provides information of early warning about cyclonic disturbances development in the north Indian Ocean. It is a developing of tropical cyclones and coastal areas likely to experience adverse weather is indicated. This is announced by Cabinet Secretary and other senior officers of Govt. of India including the Chief Secretaries of the concerned maritime states.
2. **Cyclonic alerts:** Issue at least 48 hours prior to the expected commencement of adverse weather over the coastal belts by the concerned CWC/ACWC's.
3. **Cyclone warning:** This will be issued at least 24 hours in advance. These warnings are continued to issue at 3 hourly intervals providing the latest information of cyclones positions, intensities (maximum sustained surface wind speeds), time and points of landfall together with storm surge height and type of damage expected.
4. **Post landfall scenario:** This will be announced about 12 hours before landfall and continue till such time the cyclone-force winds are expected in the interior areas. At this stage, the district collectors of all interior districts besides the coastal areas likely to be affected are included in the bulletin. After the weakening of the cyclone into a depression stage, a final message on 'de-warning' is issued to all concerned.

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